

Mathematics, 2017 EC, entrance exam, version 1

Grade 12 · Mathematics · 2017 E.C.

Time allowed: 120 minutes

Questions: 42

Total marks: 42

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following defines prime number? (1)
 - A. It is a natural number that has an odd number of factors
 - B. It is an odd natural number that has only two factors
 - C. It is a natural number that has exactly two distinct factors
 - D. It is a natural number where one of its factors is 3
2. The GCF of two numbers is 9 and their LCM is 54. If one of the numbers is 18. What is the other number? (1)
 - A. 36
 - B. 54
 - C. 27
 - D. 72
3. If $x = 0.424424442\ldots$ and $y = 0.242242224\ldots$ then the fraction form of $x + y$ is equal to (1)
 - A. $6/10$
 - B. $3/4$
 - C. $2/3$
 - D. $4/5$
4. For any real numbers a , b and c . Which one of the following is true? (1)
 - A. If $a < b$ then $a^2 < b^2$
 - B. If $a < b$ then $ac < bc$
 - C. If $a^2 < b^2$ then $b < a$
 - D. If $a < b$ and $b < c$ then $c > a$
5. Let a and b be positive integers and p and q are real numbers. Which of the following is true? (1)
 - A. $(a+b)^p = a^p + b^p$
 - B. $(a)^{-p} / (b)^{-q} = b^q / a^p$
 - C. $(a)^{-p} / (b)^{-q} = b^q / a^p$
 - D. $a^p / (a)^{-q} = (a)^{p-q}$
6. Which one of the following is the rationalizing factor to rationalize the denominator of $7/(3+\sqrt{5})$? (1)
 - A. $(\sqrt{3}+\sqrt{5})/(\sqrt{3}+\sqrt{5})$
 - B. $(\sqrt{3}-5)/(\sqrt{3}-5)$
 - C. $(\sqrt{3}-\sqrt{5})/(\sqrt{3}-\sqrt{5})$
 - D. $(3-\sqrt{5})/(3-\sqrt{5})$
7. According to division algorithm what should be the value of q and r respectively so that $290 = 11q + r$? (1)
 - A. 3 and 26
 - B. 26 and 4
 - C. 25 and 5
 - D. 24 and 6
8. Kidest bought 2 kilogram of sugar and a can of tomato and paid 100 birr. Her sister Tigist paid 150 birr for a kilogram of sugar and 3 cans of tomato. What was the price of a can of tomato? (1)
 - A. 50 birr
 - B. 30 birr
 - C. 15 birr
 - D. 40 birr
9. Let $\{ 4x - 8y = -3, 2x + ky = 16 \}$ be system. What is the value of k so that the system has no solution? (1)
 - A. $k = -4$
 - B. $k = 4$
 - C. $k \neq -4$
 - D. $k \neq 4$

10. If one root of the quadratic equation $x^2 - x + m = 0$ exceeds the other by 5. What is the value of m ? (1)
- A. -6
B. 6
C. $5/2$
D. $-3/2$
11. Let $A = \{1, 2, \{3,4\}, 5\}$ be a set. Which of the following is not true? (1)
- A. $\{\{3,4\}\} \subseteq A$
B. $\{4\} \in A$
C. $\emptyset \subseteq A$
D. $\{1,2,5\} \subset A$
12. Let $A = \{x: x^2 - 4 = 0\}$ and $B = \{x: x \text{ is a positive factor of } 4\}$. Which one of the following is $B \setminus A$? (1)
- A. $\{1,4\}$
B. $\{2,4\}$
C. $\{1,2,4\}$
D. $\{2\}$
13. For any two non empty sets A and B if $A \subset B$, then which one of the following is always true? (1)
- A. $A \cap B = B$
B. $A \cup B = A$
C. $A \Delta B = A \cup B$
D. $(A \cup B) \cap A = A$
14. In a class of 52 students 34 passed in Mathematics and 28 passed in English. If the number of students who failed in both subjects is 9. What is the number of students who passed in both subjects? (1)
- A. 19
B. 10
C. 9
D. 6
15. Which one of the following is the range of the function $f(x) = (2x+3)/(x-1)$? (1)
- A. $\mathbb{R} \setminus \{1\}$
B. $\mathbb{R} \setminus \{3\}$
C. $\mathbb{R} \setminus \{2\}$
D. $\mathbb{R}$
16. Which one of the following is true about the function $f(x) = 3x^{4/5}$? (1)
- A. The domain of f is $\mathbb{R} \setminus \{0\}$
B. The range of f is $[0, \infty)$
C. f is increasing on $(-\infty, 0]$
D. f is decreasing on $[0, \infty)$
17. What is the solution set of the equation $\operatorname{sgn}(x^2-1) = -1$? (1)
- A. $[-1, 1]$
B. $(-1, 1)$
C. $(-\infty, -1] \cup [1, \infty)$
D. $(-\infty, -1) \cup (1, \infty)$
18. Which one of the following is the solution set of the equation $1/(x-2) + 6x/(x^2-4) = 3/(x+2)$? (1)
- A. $\{-4, 2\}$
B. $\{-2\}$
C. $\{-1, 1\}$
D. $\emptyset$
19. Which of the following is the solution of the rational inequality $(x^2-6x)/(x-12) > 1$? (1)
- A. $(3, 4)$
B. $(12, \infty)$
C. $(3, 4] \cup (12, \infty)$
D. $(3, 4) \cup (12, \infty)$
20. Which one of the following function does not have a vertical asymptote? (1)
- A. $f(x) = 3/(x^2+1)$
B. $f(x) = (x+1)/(x^2-1)$
C. $f(x) = (2x+3)/(x^3+1)$
D. $f(x) = 1/x^2$

21. In a class there are 8 more boys than girls. If the ratio of boys to girls is 4 to 3. What is the number of girls in the class? (1)
- A. 32
B. 24
C. 22
D. 30
-
22. Let $A = \begin{bmatrix} 3 & a+b \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & 6 \\ 2 & a-b \end{bmatrix}$. If $A = B$, then the value of a is: (1)
- A. 2
B. 3
C. 4
D. 5
-
23. Which one of the following matrix has no equal number of rows and columns? (1)
- A. Square matrix
B. Rectangular matrix
C. Diagonal matrix
D. Identity matrix
-
24. What is the solution of the system $\{x+y=5, 2x-y=1\}$? (1)
- A. (3,2)
B. (1,4)
C. (4,1)
D. (2,3)
-
25. Let A and B be two invertible matrices of order 3. Suppose $\det(A)=8$ and $\det(B)=16$. What is $\det(A^{-1}B)$? (1)
- A. 2
B. $1/2$
C. 4
D. $1/4$
-
26. What is the value of x , so that matrix $A = \begin{bmatrix} 2 & -3 \\ x & 2 \end{bmatrix}$ does not have an inverse? (1)
- A. $4/3$
B. $-3/4$
C. $-4/3$
D. $3/4$
-
27. Which of the following relation holds for the sequence -8, -2, 4, 10, ...? (1)
- A. $a_n = a_{n-1} - 6$
B. $a_n = a_{n-1} + 6$
C. $a_n = a_{n-1} - 3$
D. $a_n = a_{n-1} - 3$
-
28. Which of the following is neither arithmetic nor geometric sequence? (1)
- A. 2,6,8,12,...
B. 5,2,-1,-4,...
C. 27,9,3,1,...
D. 0.1,0.01,0.001,...
-
29. What is the general term of the sequence A_n , if the 3rd term is 11 and the common difference is 4? (1)
- A. $A_n = 2n-3$
B. $A_n = 4n-1$
C. $A_n = 6n-3$
D. $A_n = 3n+1$
-
30. The sum of integers from 1 to 100 that are divisible by 15 is: (1)
- A. 180
B. 190
C. 300
D. 315
-
31. The sum of series $1 - 1/2 + 1/4 - 1/8 + \dots$ is equal to: (1)
- A. $3/2$
B. $1/2$
C. $2/3$
D. 2
-
32. Which one of the following is a convergent sequence? (1)
- A. $\{(5/3)^n\}$
B. $\{3n/(n+1)\}$
C. $\{n^2/(2n+1)\}$
D. $\{(1/n)^{-1}\}$

33. A man accepts a position with initial salary of Birr 12500 per year, increasing by Birr 250 each year. What will be his salary at the beginning of 12th year? (1)

A. 14,750
B. 15,000
C. 15,250
D. 14,500

34. Which of the following is the average rate of change of $f(x)=x^3-2x$ on $[1,3]$? (1)

A. 10
B. -10
C. 11
D. -11

35. Let $f(x)=\pi^2-3$, then $f'(2)$ is equal to: (1)

A. 2π
B. 4
C. -1
D. 0

36. Let $f(x)=kx^2-2x+1$ and $f'(2)=10$, then what is k ? (1)

A. 2
B. 3
C. 4
D. 5

37. If $f'(1)=5$, $f(1)=4$ and $g'(4)=5$, then $(g \circ f)'(1)$ is: (1)

A. 20
B. 25
C. 15
D. 30

38. What is the absolute minimum value of $f(x)=x^2+2x$ on $[-2,2]$? (1)

A. -3
B. 8
C. 0
D. -1

39. Which one of the following is the equation of tangent line to $f(x)=3x^2+4x-5$ at $(0,-5)$? (1)

A. $y=4x-5$
B. $y=4x+5$
C. $y=-4x-5$
D. $y=-4x+5$

40. The volume of spherical balloon is increasing at rate $24 \text{ cm}^3/\text{sec}$. What is rate of change of surface area when radius=8 cm? (1)

A. $6 \text{ cm}^2/\text{sec}$
B. $8 \text{ cm}^2/\text{sec}$
C. $4 \text{ cm}^2/\text{sec}$
D. $3 \text{ cm}^2/\text{sec}$

41. What is interquartile range (IQR) in statistics? (1)

A. Difference between mean and median
B. Difference between 3rd quartile and 1st quartile
C. Difference between maximum and minimum
D. Half of difference between 3rd and 1st quartile

42. What is the mean deviation from the mean for data: 5,3,7,8,4,9? (1)

A. 2
B. 4
C. 3
D. 0